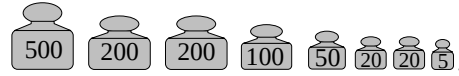


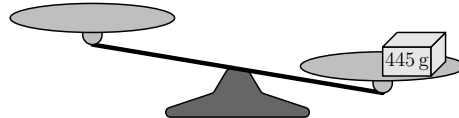
GRADES 5 AND 6

SAMPLE QUESTION FOR 3 POINTS

Peter has a package weighing 445 g and the following eight weights:



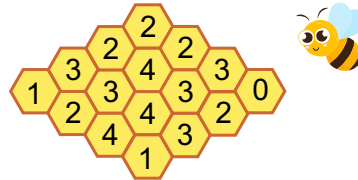
He puts the package on a scale, as shown. What is the smallest number of weights he needs to balance the scale?



- (A) 2 (B) 3 (C) 4 (D) 5 (E) 6

SAMPLE QUESTION FOR 4 POINTS

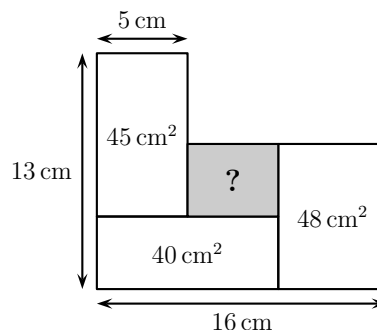
The figure below shows a honeycomb with 16 cells. Some of the cells contain honey. The number in each cell indicates how many of its neighboring cells contain honey. Two cells are neighbors if they share a common edge. How many cells in the honeycomb contain honey?



- (A) 7 (B) 8 (C) 9 (D) 10 (E) 11

SAMPLE QUESTION FOR 5 POINTS

The diagram shows four touching rectangles. What is the area of the shaded rectangle?



- (A) 12 cm^2 (B) 14 cm^2 (C) 16 cm^2 (D) 18 cm^2 (E) 20 cm^2

GRADES 7 AND 8

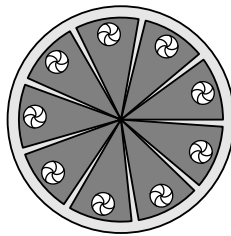
SAMPLE QUESTION FOR 3 POINTS

What is the value of $\frac{20 \times 24}{2 \times 0 + 2 \times 4}$?

- (A) 12 (B) 30 (C) 48 (D) 60 (E) 120

SAMPLE QUESTION FOR 4 POINTS

Carolina baked a cake and cut it into ten equal pieces. She ate one piece and then arranged the remaining pieces evenly, as shown. What is the measure of the angle between any two adjacent pieces?



- (A) 5° (B) 4° (C) 3° (D) 2° (E) 1°

SAMPLE QUESTION FOR 5 POINTS

Captain Flint asked four of his pirates to write on a piece of paper how many gold, silver, and bronze coins were in the treasure chest. Their responses are shown in the diagram but unfortunately part of the paper was damaged. Only one of the four pirates told the truth. The other three lied in all their answers. The total number of coins is 30. Who told the truth?

	Gold	Silver	Bronze
Tom		9	11
Al	7		12
Pit	10		10
Jim	9	10	

- (A) Tom (B) Al (C) Pit (D) Jim
(E) We cannot be sure.